Final System Verification

1. Overview

The SYS_TB testbench runs 9 test suites. Each suite is a full set of test cases for one UART configuration, and the DUT is reset before every suite. The suites cover three prescale values (32, 16, 8) and three parity modes (even, odd, disabled).

PRESCALE	EVEN PARITY	ODD PARITY	PARITY DISABLED
32	Suite 1	Suite 2	Suite 3
16	Suite 4	Suite 5	Suite 6
8	Suite 7	Suite 8	Suite 9

2. Test Cases per Suite

	TEST GROUP	CASES	DESCRIPTION
1	Read config / divider registers	2	Read REG2 (UART config) and REG3 (TX divider ratio) and compare with the reference model
2	Register read operations	6	Read addresses 4, 5, 6, 7, 8 and 15 after they are written
3	ALU with operands (command 0xCC)	11	ADD, SUB, MUL, DIV, AND, OR, XOR, compare equal, compare greater, shift right, shift left
4	ALU without operands (command 0xDD)	4	ADD, MUL, NAND, shift right, using the values stored in REG0 and REG1
5	No false error flags	1	Verify that parity_error and framing_error stay low during valid traffic
6	Parity error injection	1	Send a frame with a wrong parity bit and verify that parity_error is asserted (only when parity is enabled)
7	Framing error injection	1	Send a frame with stop bit = 0 and verify that framing_error is asserted
8	Recovery after error frames	1	Read address 4 after the error frames to confirm the system still works

Stimulus (not counted as separate test cases):

- UART configuration write to REG2.
- Register writes to addresses 4, 5, 6, 7, 8 and 15, which are checked by the read operations that follow them.

3. Total Test Case Count

SUITE TYPE	CASES PER SUITE	SUITES	TOTAL
PARITY ENABLED (EVEN / ODD)	27	6	162
PARITY DISABLED	26	3	78
TOTAL		9	240

4. Checking Methodology

- Output checking: The testbench waits for the UART TX to start, with a timeout of 80 TX clock periods. It then samples UART_TX_OUT and compares the received frame (start bit, data, parity bit if enabled, stop bit) with the expected frame using ===.
- Reference models: RF_MODEL is the register file model, and the function ALU_LOW computes the expected ALU result.
- Error flag monitoring: Event monitors on the rising edges of parity_error and framing_error record whether each flag was seen.
- Final summary: The testbench prints the total, passed and failed counts, followed by ALL TESTS PASSED or SOME TESTS FAILED.

5. Results

TOTAL	PASSED	FAILED	FINAL RESULT
240 (EXPECTED)	240	0	ALL TESTS PASSED

```
# ============================================================
# SIMULATION FINISHED
# ============================================================
# TOTAL TEST CASES  = 240
# PASSED            = 240
# FAILED            = 0
# ============================================================
# RESULT : ALL TESTS PASSED
# ============================================================
# ** Note: $stop    : sys_tb.v(139)
#    Time: 124710820865 ps  Iteration: 0  Instance: /SYS_TB
# Break in Module SYS_TB at sys_tb.v line 139
```